

AWS Generative AI

Quick Interview Revision Guide

Goal: Basic exposure level understanding of Amazon Bedrock, Amazon Q Developer, RAG, AI Agents, and SageMaker for interviews.

1. Amazon Bedrock

- Fully managed AWS service for Generative AI.
- Provides access to Foundation Models (FMs) from providers like Anthropic Claude, Meta Llama, Amazon Nova, and others.
- No need to manage infrastructure or train models from scratch.
- Use Cases: Chatbots, content generation, summarization, Q&A, code assistance.
- Key Features: Model access, playgrounds, APIs, knowledge bases, agents.
- **Interview Answer:** Bedrock allows organizations to build Generative AI applications using pre-trained foundation models through a managed AWS service.

2. Amazon Q Developer

- AI-powered assistant for developers and cloud engineers.
- Integrates with IDEs such as VS Code.
- Helps generate code, explain code, troubleshoot issues, and answer AWS-related questions.
- Improves developer productivity.
- **Interview Answer:** Amazon Q Developer is an AI assistant that helps developers build, understand, and operate applications on AWS.

3. RAG (Retrieval-Augmented Generation)

- Technique that improves LLM responses using external knowledge sources.
- Instead of relying only on model training data, relevant information is retrieved before generating an answer.
- Components: User Query → Retrieval → Context → LLM Response.
- Uses embeddings and vector databases.
- Reduces hallucinations and improves accuracy.
- **Interview Answer:** RAG combines information retrieval with LLMs to generate more accurate and context-aware responses.

4. AI Agents in Bedrock

- AI agents can reason, plan, and perform tasks.
- Agents can use tools, APIs, databases, and knowledge bases.
- Often combine LLMs with RAG.
- Can automate workflows and business processes.
- **Interview Answer:** Bedrock Agents help automate multi-step tasks by allowing AI models to use tools and external systems.

5. Amazon SageMaker

- AWS platform for building, training, and deploying machine learning models.
- Supports the full ML lifecycle.
- Key Concepts: Training, Deployment, Endpoints, MLOps.
- Used when organizations want to create or train their own ML models.
- **Interview Answer:** SageMaker is AWS's machine learning platform used to build, train, deploy, and manage ML models.

6. Bedrock vs SageMaker

- Amazon Bedrock: Use pre-trained foundation models.
- Amazon SageMaker: Build, train, and deploy custom ML models.
- Bedrock is simpler for Generative AI applications.
- SageMaker provides deeper ML customization and control.

7. Quick Interview Questions

- What is Amazon Bedrock?
- What is Amazon Q Developer?
- What is RAG and why is it important?
- How do AI Agents work?
- What is Amazon SageMaker?
- What is the difference between Bedrock and SageMaker?
- How does RAG reduce hallucinations?
- What are embeddings?

8. 30-Second Interview Summary

- I explored AWS Generative AI services including Amazon Bedrock, Amazon Q Developer, RAG concepts, AI Agents, and Amazon SageMaker. I understand their purpose, common use cases, and how they fit into AWS AI solutions. Bedrock provides access to foundation models, Amazon Q assists developers, RAG improves response accuracy using external knowledge, AI Agents automate tasks, and SageMaker supports machine learning model development and deployment.

